HOMEWORK ASSIGNMENT 3

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

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Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
  - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
- Solutions to this assignment will be released at 8:00am on Friday, September 11 in time for the corresponding quiz on **Monday September 14**.
  - **Kindly Note:** Quiz 03 will be the full 15 questions and last for the full 20 minutes. 10 questions will be based off this week's material; 5 questions will be based off last week's material (to compensate for the fact that Quiz 02 did not take place as scheduled)

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1. **Is This Die Balanced?** I have recently acquired a six-sided die that, when rolled 12 times, landed on an even number (i.e. 2, 4, or 6) ten times. With this information, I'd like to investigate whether or not the die is balanced (has an equal chance of landing on an even or odd number) or loaded in favor of even numbers (has a higher chance of landing on an odd number than an even one).

   (a) Describe an observational unit in the context of this dataset.

   (b) Define the parameter of interest.

   (c) Calculate the value of the statistic.

   (d) Consider the variable: "whether a particular roll landed on an even number or not." What are the possible levels of this variable, and which level are you considering a "success?"

   (e) State the research question in two ways: one verbally (using words and only minimal math), and one mathematically (involving the parameter of interest you defined above).

   (f) Fill in the entries in the right-most column of the table below, to set up the chance model for this study.

One Coin Flip	
Coin Landing Heads	
Tails	
Chance of Heads	
One Repetition	

   (g) Use the One Proportion Applet to simulate 2,000 repetitions of the chance model. Include a screenshot or sketch of your plot below. **Don't close out of the applet yet - the next question will ask you to modify your plot!**

(h) Provide an approximate value for the probability that ten out of twelve rolls of a balanced six-sided die land on an even number. Do this by:

1) Navigating back to the applet, and find the line that says "As extreme as."

2) Clicking the $\geq$ symbol twice until it turns into the $=$ symbol, and entering the appropriate value (it's up to you to identify the correct value to put based on the problem statement!) in the provided blank.

Again, do not close out of the applet as we will modify it further in the next part of this question.

(i) Explain why the value you obtained in part (g) is just an *approximation* to the desired probability.

(j) Provide an approximate value for the probability that ten or more rolls out of a total of twelve rolls of a balanced six-sided die land on an even number. Do this by:

1) Navigating back to the applet, and going back to the line that says "As extreme as."

2) Clicking the $=$ symbol until it turns into the $\geq$ symbol (make sure you still have the value you entered before in the provided blank space).

(k) The true/exact probability that ten or more out of twelve rolls of a balanced six-sided die land on an even number can be computed (using mathematical tools that are outside the scope of this course) to be 0.0192. Provide an interpretation of this value.

(l) Is it likely that ten or more of twelve rolls of a balanced six-sided die land on an even number? What does this mean about the evidence for/against the possibility that the die is balanced, vs. the possibility that the die is biased in favor of even numbers?

2. **Cat Loafs.** The term "loafing" is commonly used among cat owners to describe the act of a cat sitting in a particular way. When "loafed," a cat can either curl their tail clockwise or counterclockwise. Suppose our research question is: are cats more likely to curl their tails clockwise when loafing? I (Dr. Ethan) conducted a (very rudimentary) study in which I watched a few videos on *YouTube* of cats loafing; among the 23 loafing cats I observed, 14 of them curled their tails clockwise (and the remainder curled their tails counterclockwise). The particular research conjecture we'll consider is: do cats prefer to curl their tails clockwise when loafed?

(a) Define the parameter of interest.

(b) Calculate the value of the statistic

(c) Fill in the entries in the right-most column of the table below, to set up the chance model for this study.

One Coin Flip	
Coin Landing Heads	
Tails	
Chance of Heads	
One Repetition	

(d) Shown below is a plot from the One Proportion Applet representing 1,000 repetitions of the chance model. Also indicated in the plot is the fact that around 20% of the 1,000 repetitions corresponded to samples with sample proportions of 14/23 or greater.

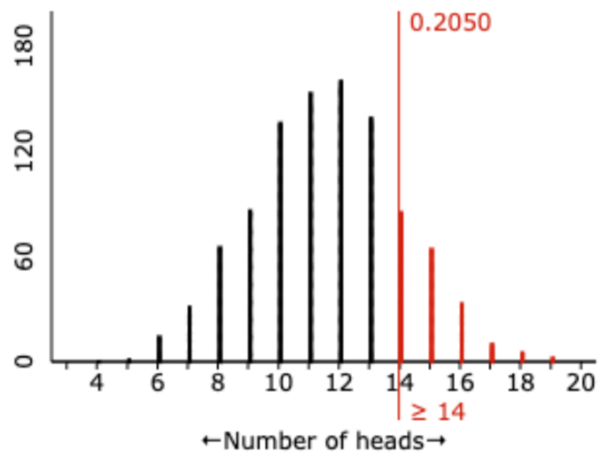

Figure 4: 1,000 repetitions, as conducted by the One Proportion applet.

This shows that it is not that unlikely to observe a sample proportion of 14/23 or greater. Does this **prove** that cats prefer to curl their tails clockwise when loafed? If not, how would you phrase the conclusions of the study, based on the simulation of the chance model?